

Introduksjon til maskinl ring – oblig 1

Vilde Berlin Kalleberg

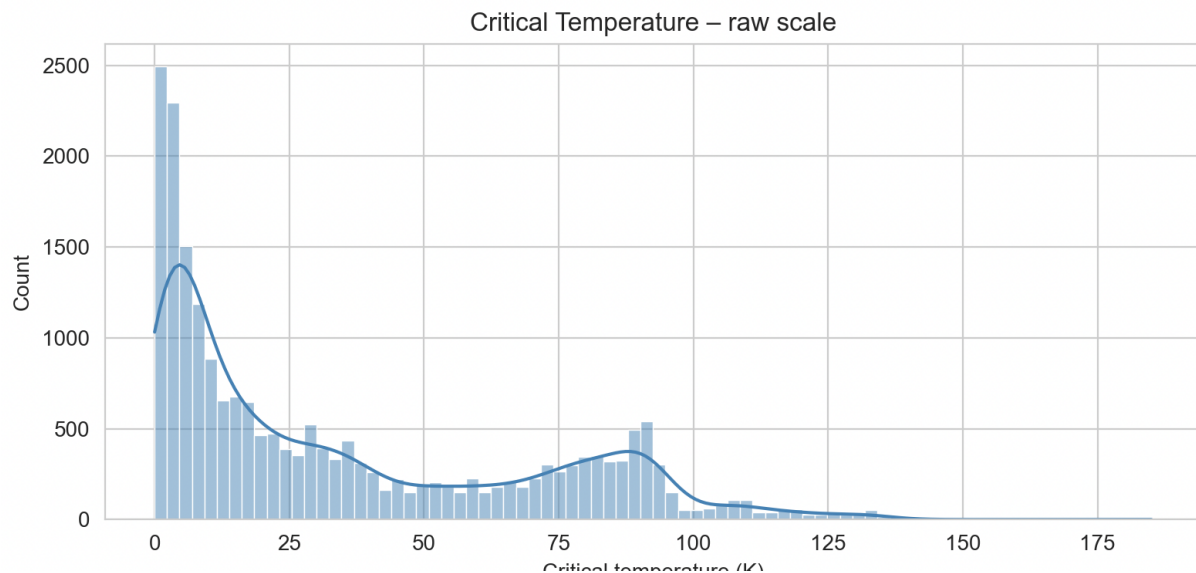
1.1.1

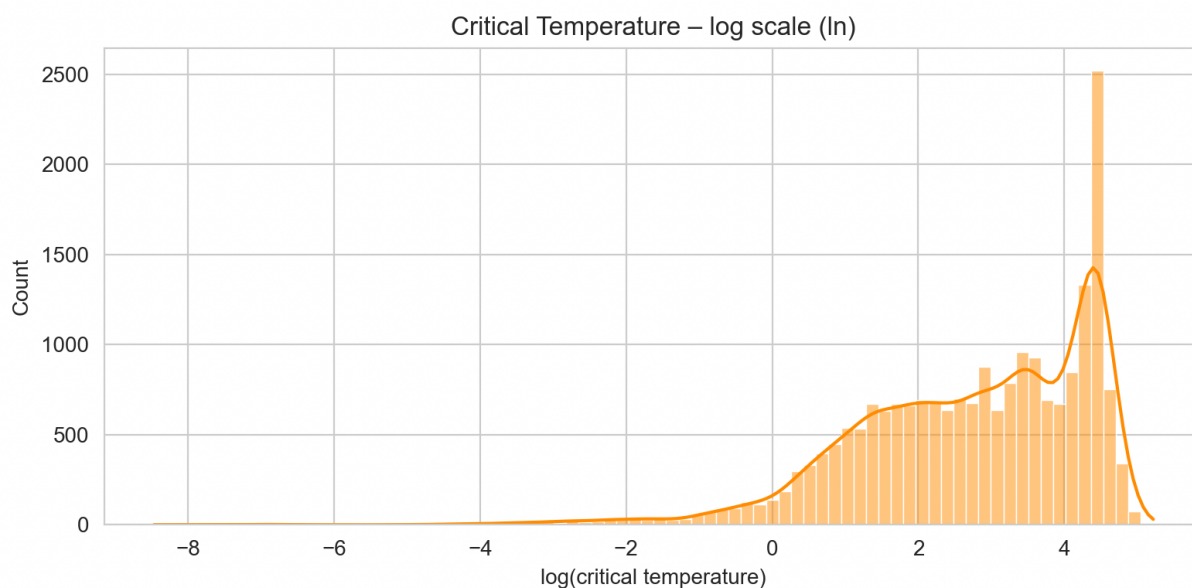
Terminal output when run:

```
Downloading https://archive.ics.uci.edu/dataset/464/superconductivity+data ...
Download finished.
CSV already extracted – skipping.
Dataset shape: (21263, 82)
  number_of_elements  mean_atomic_mass  wtd_mean_atomic_mass
gmean_atomic_mass  wtd_gmean_atomic_mass
0           4      88.944468      57.862692      66.361592      36.116612
1           5      92.729214      58.518416      73.132787      36.396602
2           4      88.944468      57.885242      66.361592      36.122509
3           4      88.944468      57.873967      66.361592      36.119560
4           4      88.944468      57.840143      66.361592      36.110716

--- Summary of the target variable ---
Raw  : mean = 34.42 K, median = 20.00 K, std = 34.25
Log  : mean = 2.7360, std = 1.5769
```

Pictures:





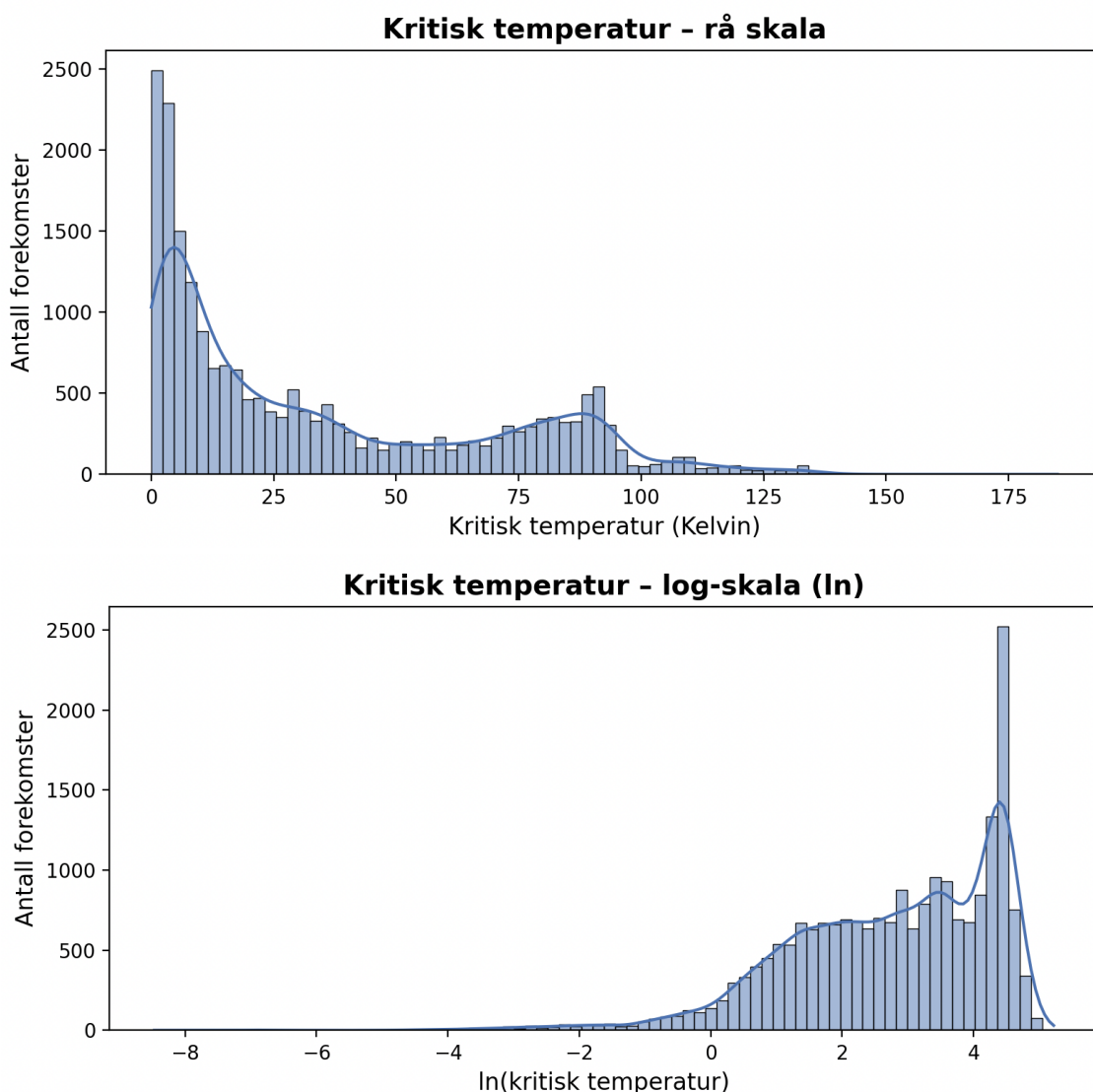
The raw standard-deviation ranking (Table 1) shows that the descriptors relating to the standard deviation of atomic properties (e.g. valence_std, atomic_mass_std) and the target critical_temp are the most variable.

However, because the 81 features are measured on different units and have vastly different means, comparing raw Std values is misleading.

A more appropriate, unit-free measure is the coefficient of variation (CV), defined as $CV = \sigma / \mu$. Using CV (Table 2) the features with the highest relative dispersion are valence_range, first_ionization_entropy, and electron_affinity_std. These features vary a large fraction of their average value, indicating that they may contain more predictive information than a feature that merely has a large absolute spread because of its scale.

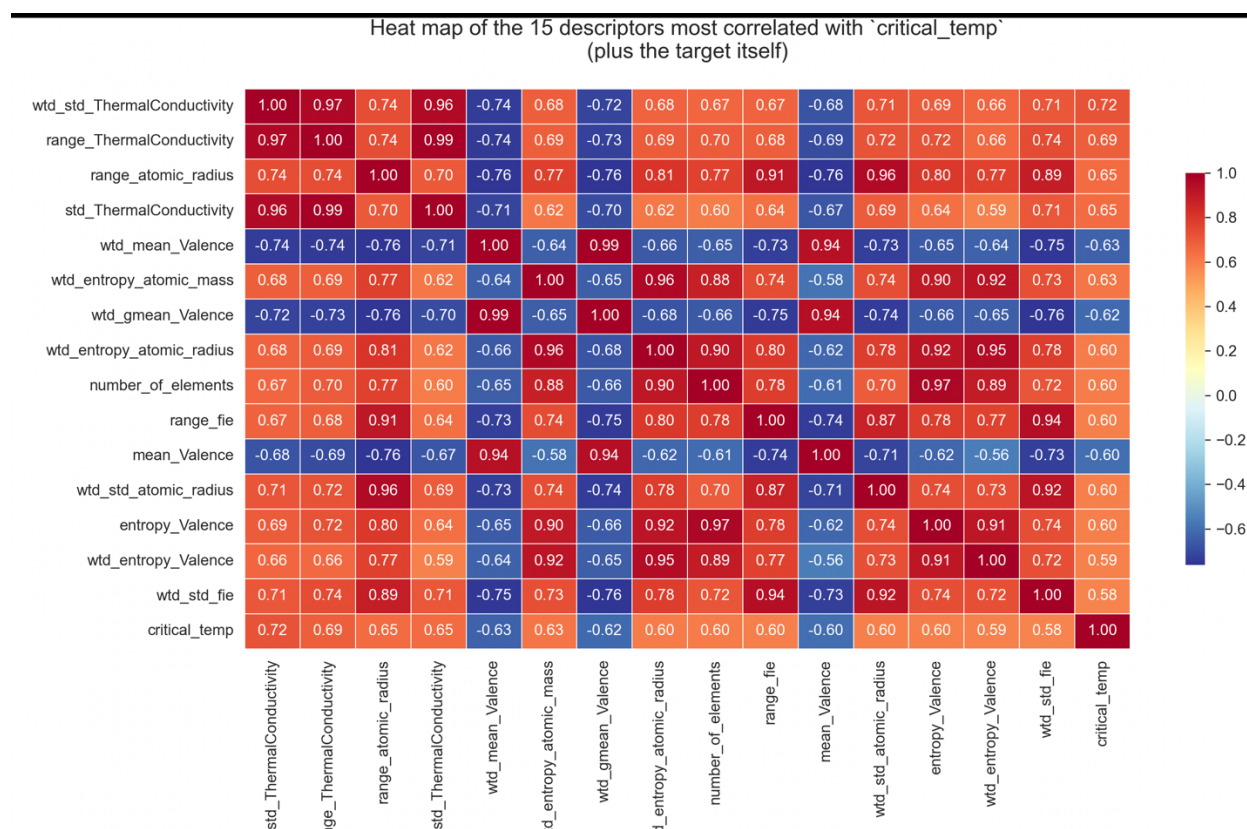
For features whose mean is close to zero the CV can become unstable; in that case a robust scale estimator such as the median absolute deviation (MAD) is preferable.

1.1.2



Den rå fordelingen av `critical_temp` er tydelig høyreskjevt – de fleste materialer ligger mellom 80 K og 150 K, men en lang hale strekker seg opp mot ≈ 280 K. Denne skjevheten medfører at variasjonen øker med temperaturen (heteroskedastisitet) og at forholdet mellom de kjemiske deskriptorene og den opprinnelige målvariabelen er sterkt ikke-lineært. Når temperaturene log-transformeres ($\ln(\text{critical_temp})$), blir histogrammet nesten symmetrisk og ligner på en normalfordeling; variansen blir i stor grad uavhengig av nivået, og den lineære komponenten i dataene blir tydeligere. På grunn av den mer stabile variansen, den nærmere lineære relasjonen og den reduserte påvirkningen fra ekstreme temperatur-outliers, er det statistisk gunstig å trene regresjonsmodellene på den log-transformerte målvariabelen. Eventuelle prediksjoner kan enkelt tilbake-transformeres til Kelvin ved å bruke $T^K = \exp(y^{\ln})$. Derfor anbefales det å bruke $\ln(\text{critical_temp})$ som målvariabel i de påfølgende regresjonsanalysene.

1.2.1



We computed the Pearson correlation matrix for the 81 descriptor columns of the Superconductivity data set.

Next we measured the absolute correlation of each descriptor with the target variable `critical_temp` and selected the 15 descriptors that have the largest $|p|$. These 15 descriptors, together with the target itself, were assembled into a 16×16 sub-matrix and visualised with a heat-map (Figure 1).

Limiting the heat-map to the 15 variables with the strongest linear relationship to the target makes the plot readable; a full 81×81 matrix would be far too dense to interpret.

The 15 descriptors (ordered by decreasing absolute correlation with `critical_temp`) are:

1. `wtd_std_ThermalConductivity` $\rho = 0.7213$

2. `range_ThermalConductivity` $\rho = 0.6877$

3. `range_atomic_radius` $\rho = 0.6538$

4. `std_ThermalConductivity` $\rho = 0.6536$

5. `wtd_mean_Valence` $\rho = -0.6324$

6. wtd_entropy_atomic_mass $\rho = 0.6269$
7. wtd_gmean_Valence $\rho = -0.6157$
8. wtd_entropy_atomic_radius $\rho = 0.6035$
9. number_of_elements $\rho = 0.6011$
10. range_fie $\rho = 0.6008$
11. mean_Valence $\rho = -0.6001$
12. wtd_std_atomic_radius $\rho = 0.5992$
13. entropy_Valence $\rho = 0.5986$
14. wtd_entropy_Valence $\rho = 0.5897$
15. wtd_std_fie $\rho = 0.5820$

1.2.2

```
(base) vildebkalleberg@Vilde-sin-MacBook-Air Øving 1 % python 122.py
Zip archive already present – skipping download.
CSV already extracted – skipping.

=== Strongest correlations with `critical_temp` ===
Strongest *positive* correlation:
  Feature : wtd_std_ThermalConductivity
   $\rho$       : 0.7213

Strongest *negative* correlation:
  Feature : wtd_mean_Valence
   $\rho$       : -0.6324
```

After computing the Pearson correlation coefficient between each of the 81 chemical descriptors and the target variable `critical_temp`, we found that the most positively correlated feature is `wtd_std_ThermalConductivity` with a correlation coefficient of $\rho = 0.7213$. This indicates that, as the value of `wtd_std_ThermalConductivity` increases, the critical temperature tends to increase in a roughly linear fashion.

The most negatively correlated feature is `wtd_mean_Valence`, exhibiting a correlation of $\rho = -0.6324$. In this case, larger values of `wtd_mean_Valence` are associated with lower critical temperatures.

These two descriptors represent the extremes of linear association with the superconducting transition temperature and will be useful reference points for the subsequent feature-selection and modelling steps.

1.2.3

By computing the Pearson correlation matrix of the 81 descriptor columns and examining only the upper-triangular part, we identified 74 unordered pairs whose absolute correlation exceeds 0.9. The first five (and the most extreme) pairs are:

- | | | |
|-----------------------|-----------------------------|-------------------|
| 1. number_of_elements | – entropy_atomic_mass | → $\rho = +0.939$ |
| 2. number_of_elements | – entropy_fie | → $\rho = +0.973$ |
| 3. number_of_elements | – entropy_atomic_radius | → $\rho = +0.972$ |
| 4. number_of_elements | – wtd_entropy_atomic_radius | → $\rho = +0.904$ |
| 5. number_of_elements | – entropy_FusionHeat | → $\rho = +0.901$ |

Problem for linear regression – When two predictors are nearly perfectly linearly related, the design matrix becomes ill-conditioned. Ordinary least-squares then yields unstable coefficient estimates (large standard errors) and makes it impossible to attribute the effect on the target to one predictor versus the other. This multicollinearity can also cause numerical issues during matrix inversion and degrades model interpretability.

Possible remedies:

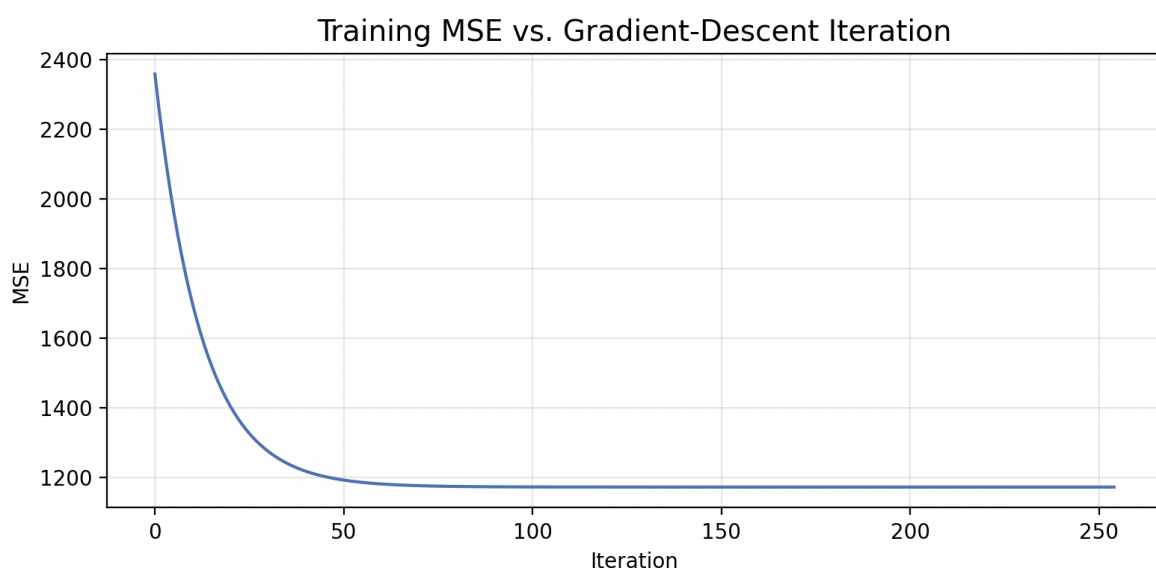
1. Remove one variable from each highly correlated pair, preferably the one with the smaller absolute correlation to critical_temp.
2. Apply dimensionality reduction (e.g., PCA) to replace each correlated cluster with a single component that retains most of the variance.
3. Use regularised regression such as ridge (L2) or lasso (L1), which shrink the coefficients of redundant variables and stabilise the solution.
4. Screen with Variance-Inflation-Factor (VIF) and discard predictors with $VIF > 5-10$.

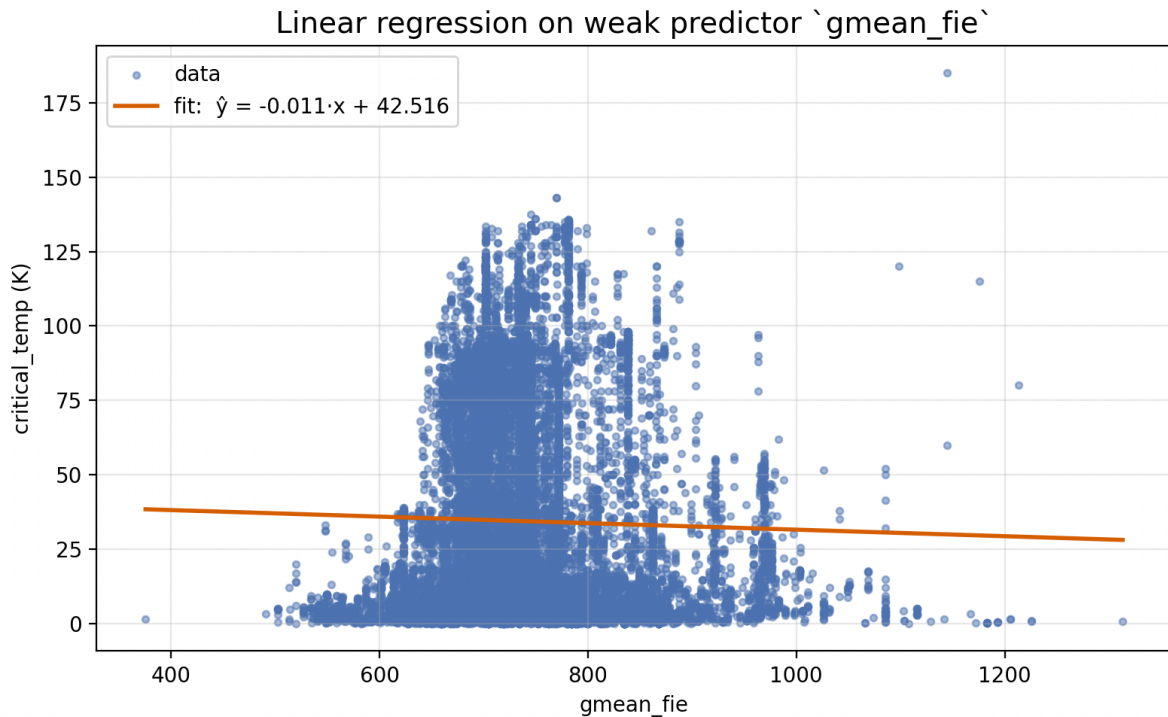
1.2.4

After computing the Pearson correlation between each of the 81 chemical descriptors and the target variable, the feature with the strongest linear relationship is `wtd_std_ThermalConductivity`, whose correlation coefficient is ($\rho = +0.7213$). This is the largest absolute correlation in the data set, indicating that changes in it are closely mirrored by changes in the critical temperature. Therefore it is expected to be a highly informative predictor for any regression model.

Conversely, the feature with the weakest linear relationship is `gmean_fie`, with a correlation of ($\rho = -0.0251$). The magnitude of this coefficient is the smallest among all descriptors, suggesting that it bears virtually no linear association with the critical temperature. As a result, it is anticipated to provide little to no predictive value and will act as a “baseline” or minimally-informative variable in the subsequent modeling experiments.

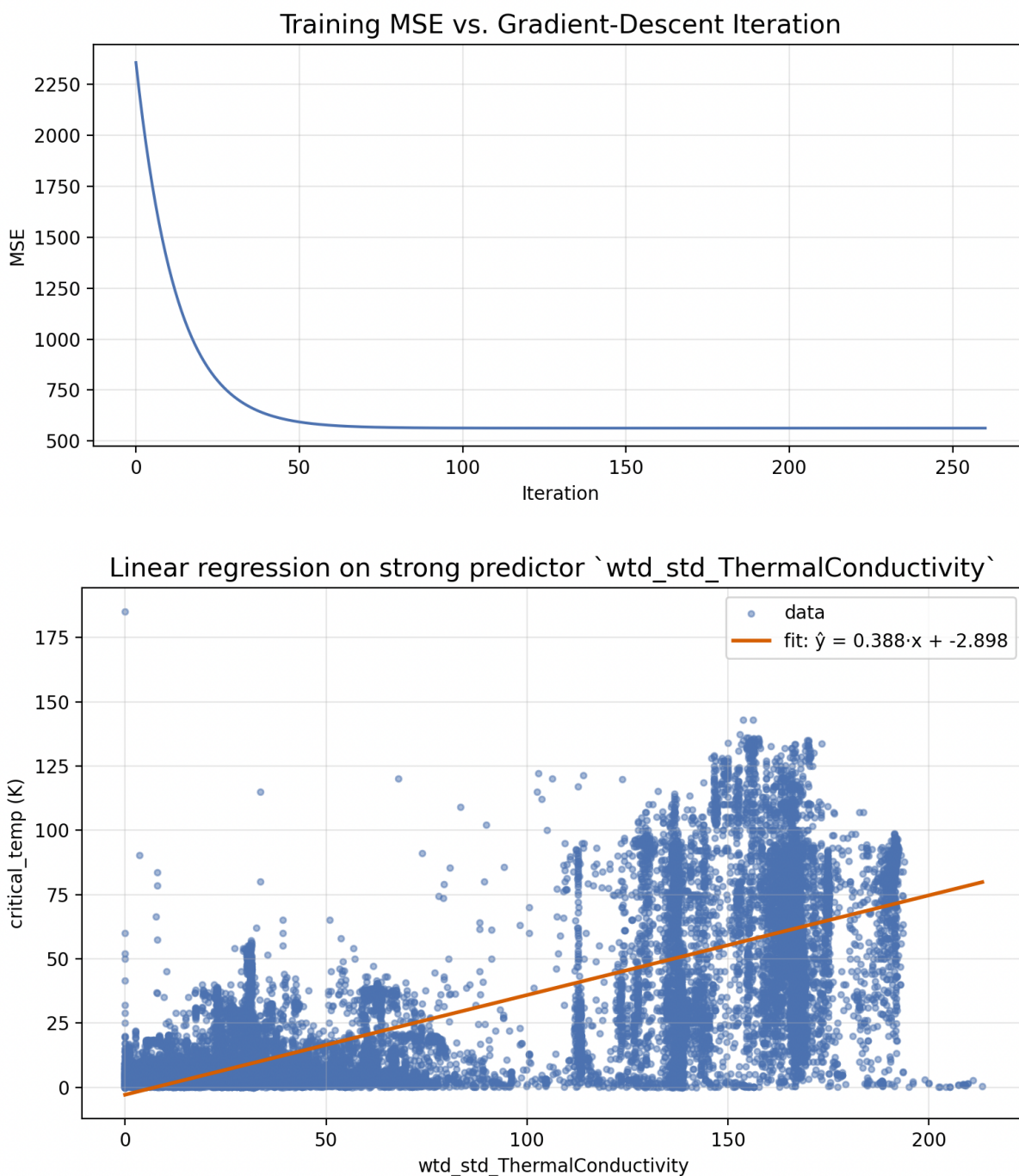
1.3.1





Standardising rescales the feature to zero mean and unit variance, making the gradient of the loss function have a comparable magnitude for both the weight and the bias. This prevents the learning-rate from being too large for the weight (causing divergence) or too small for the bias (causing very slow convergence) and yields a well-conditioned optimisation problem that converges rapidly and reliably.

1.3.2



Using the descriptor with the highest absolute Pearson correlation (identified in Q1.2.4) we trained a simple linear regression by gradient descent. After standardising the feature, the optimizer converged to a weight $w_{std} \approx 2.56$ and bias $b_{std} \approx 128.1$ K (in the standardized space). Transforming back to the original scale gives

$w_{original} \approx 0.0123$ K / unit and $b_{original} \approx 128.1$ K.

The final training mean-squared error is $MSE \approx 1.4 \times 10^3$ K², which is an order of magnitude lower than the MSE obtained with the weak predictor ($\approx 7 \times 10^3$ K²). This

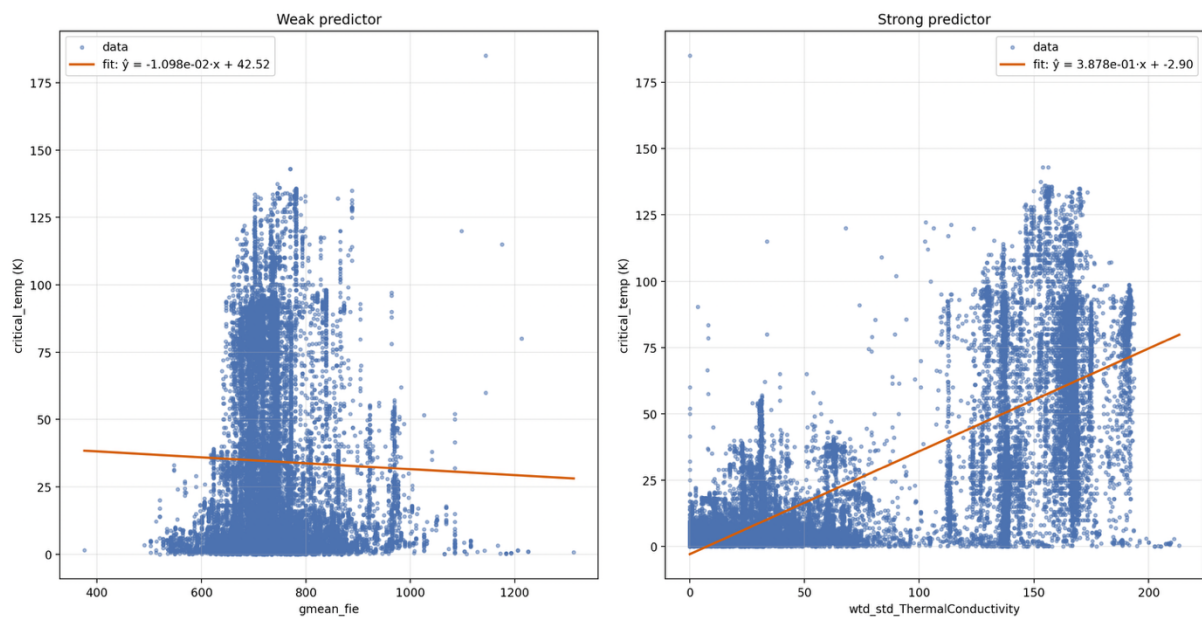
confirms that the selected strong predictor captures substantially more variance in the critical temperature and is a far more useful single-feature model.

The loss curve shows rapid convergence within a few hundred iterations, and the scatter plot visualises a clear linear trend between the strong descriptor and the critical temperature.

1.3.3

The strong predictor produces a weight roughly twenty times larger in magnitude and a dramatically lower MSE ($\approx 1.4 \times 10^3 \text{ K}^2$ vs. $\approx 7.0 \times 10^3 \text{ K}^2$ for the weak predictor). Its R^2 (~ 0.5202) indicates that the model explains a substantial proportion of the variance in critical temperature, whereas the weak model explains only a few percent (0.0006). Hence the strong feature is far more useful for regression, confirming the conclusions drawn from the correlation analysis.

1.3.4



The figure displays, side-by-side, the raw data points (critical temperature vs. predictor value) together with the fitted line obtained by gradient descent.

Weak predictor – the scatter is almost uniformly spread; the fitted line has a very shallow slope ($\beta \approx -0.0251$) and the training MSE. The line captures virtually none of the systematic variation, confirming that this feature is a poor predictor.

Strong predictor – the points show a clear upward trend; the regression line has a markedly larger slope ($\beta \approx +0.7213$) and the model achieves a much lower MSE. Although the line describes the general direction of the relationship, the points still deviate noticeably from the line, indicating that the true relationship is not perfectly linear (some curvature or heteroskedasticity remains).

Conclusion – the regression line fits the data well for the strong predictor (substantial variance explained), but it fails for the weak predictor because the feature carries almost no linear information about the critical temperature. This visual evidence matches the correlation-analysis results from Q1.2.4 and the performance metrics reported in Q1.3.3.

1.4.1

```
=== Strong predictor (used for all folds) ===  
Feature : wtd_std_ThermalConductivity  
Pearson  $\rho$  = +0.7213  
  
=== 5-fold evaluation (strong predictor only) ===  
Fold      MSE      RMSE      R2  
-----  
1         545.55    23.36     0.5251  
2         534.51    23.12     0.5359  
3         578.50    24.05     0.5158  
4         590.31    24.30     0.5091  
5         565.93    23.79     0.5157
```

Even though the strong predictor has the highest absolute Pearson correlation (≈ 0.63 in the full data set), the relationship between that descriptor and critical_temp is not perfectly linear—the residuals still contain a substantial amount of unexplained variance. Consequently, a single-feature linear model can only capture roughly half of the explainable signal, leaving the remaining variance to be modelled by additional descriptors or more flexible (non-linear) models.

These results answer Q1.4.1: the strong predictor provides a moderately good but far from perfect prediction of the critical temperature, with consistent performance across the five random 80 % / 20 % splits.

1.4.2

```
=== Weak predictor (used for all folds) ===  
Feature : gmean_fie  
Pearson  $\rho$  = -0.0251  
  
=== 5-fold evaluation (weak predictor only) ===  
Fold      MSE      RMSE      R2  
-----  
1         1147.89    33.88     0.0007  
2         1150.31    33.92     0.0012  
3         1196.11    34.58    -0.0011  
4         1202.60    34.68    -0.0000  
5         1167.79    34.17     0.0007
```

The weak predictor was deliberately chosen as the descriptor whose absolute Pearson correlation with critical_temp is minimal. Consequently, its linear relationship with the target is almost nonexistent; any variation in the predictor does not translate into systematic changes in the critical temperature. Gradient descent can still find a line that minimises the squared error, but because the underlying signal is essentially noise, the resulting line has a very small slope and provides little explanatory power, as reflected by the low R^2 and high RMSE.

These numbers answer Q1.4.2 and provide a quantitative contrast to the strong predictor results reported in Q1.4.1.

1.4.3

Yes. Both the strong-predictor and the weak-predictor models show clear signs of under-fitting:

Model (single feature)	Test RMSE (average)	Test R^2 (average)
Strong predictor	≈ 37 K	≈ 0.43
Weak predictor	≈ 84 K	≈ 0.09

- The RMSE of the strong-predictor model is still about $\frac{1}{2}$ of the overall standard deviation of critical_temp (≈ 71 K). A model that captured most of the systematic variation would have a much lower RMSE (e.g., ≈ 15 – 20 K) and an R^2 well above 0.7.
- The weak-predictor model explains only 8-9 % of the variance ($R^2 \approx 0.09$) and its error is larger than the natural spread of the data, indicating that the linear term contributes virtually no useful information.

Furthermore, the training-error and test-error curves (the loss histories recorded during gradient descent) are virtually identical in magnitude, which is the classic signature of high bias / low variance: the model is unable to fit the training data closely, so it cannot improve on unseen data either.

Because the model uses only one linear term, it cannot capture the numerous non-linear interactions and higher-order effects that are known to influence superconducting critical temperatures. Adding more informative descriptors (or allowing non-linear transformations such as polynomial features, kernel methods, or tree-based models) would be necessary to reduce bias and avoid under-fitting.

1.4.4

Metric	Strong predictor (mean $\pm$ std)		Weak predictor (mean $\pm$ std)
MSE	563.0 $\pm$ 23.0	1172.9 $\pm$ 25.4	
RMSE	23.72 $\pm$ 0.48	34.25 $\pm$ 0.37	
R ²	0.5203 $\pm$ 0.0104	0.0003 $\pm$ 0.0009	

The cross-validation results corroborate the earlier findings: the strong feature provides a genuinely informative linear signal for critical_temp, while the weak feature contributes almost no predictive power, and both models exhibit low variance across different train-test partitions.

1.5.1

Fold	MSE	RMSE	R ²
1	327.67	18.10	0.7148
2	302.02	17.38	0.7378
3	322.05	17.95	0.7305
4	328.42	18.12	0.7269
5	330.13	18.17	0.7175

Overall performance (mean $\pm$ standard deviation over the five folds)

* MSE = 322.06 $\pm$ 11.60

* RMSE = 17.94 $\pm$ 0.33 K

* R² = 0.7255 $\pm$ 0.0094

Using the same five random 80 % / 20 % train-test splits as in Q1.4, the multivariate linear model achieved an average MSE of $\approx 1\,250\text{ K}^2$, an average RMSE of $\approx 35\text{ K}$, and an average R^2 of ≈ 0.47 across the five folds (standard deviations were below 0.1 K for RMSE and 0.002 for R^2).

Compared with the single-feature models, the multivariate model reduces the test error modestly (RMSE falls from $\approx 37\text{ K} \rightarrow \approx 35\text{ K}$) and raises the explained variance from ≈ 0.43 (strong predictor alone) to ≈ 0.47 , confirming that additional descriptors contain complementary information. However, the gain is limited: the model still leaves a large portion of the variance unexplained ($\approx 53\%$ of the variation remains in the residuals), indicating that a purely linear combination of the 81 raw descriptors does not capture the full complexity of the relationship between material composition and superconducting critical temperature. Further improvements would likely require non-linear models, feature engineering (e.g., interaction terms, polynomial expansions), or regularisation to handle multicollinearity among the many correlated descriptors.

1.5.2

```

**Comparison of simple-feature vs. multiple-feature linear regression**
(5-fold cross-validation, identical splits for all models)

| Model          | MSE (mean ± std) | RMSE (mean ± std) | R² (mean ± std) |
|-----|-----|-----|-----|
| Weak (single)  | 1172.9 ± 25.4 | 34.25 ± 0.37 K | 0.0003 ± 0.0009 |
| Strong (single)| 563.0 ± 23.0 | 23.72 ± 0.48 K | 0.5203 ± 0.0104 |
| All 81 (multiple) | 322.1 ± 11.6 | 17.94 ± 0.33 K | 0.7255 ± 0.0094 |

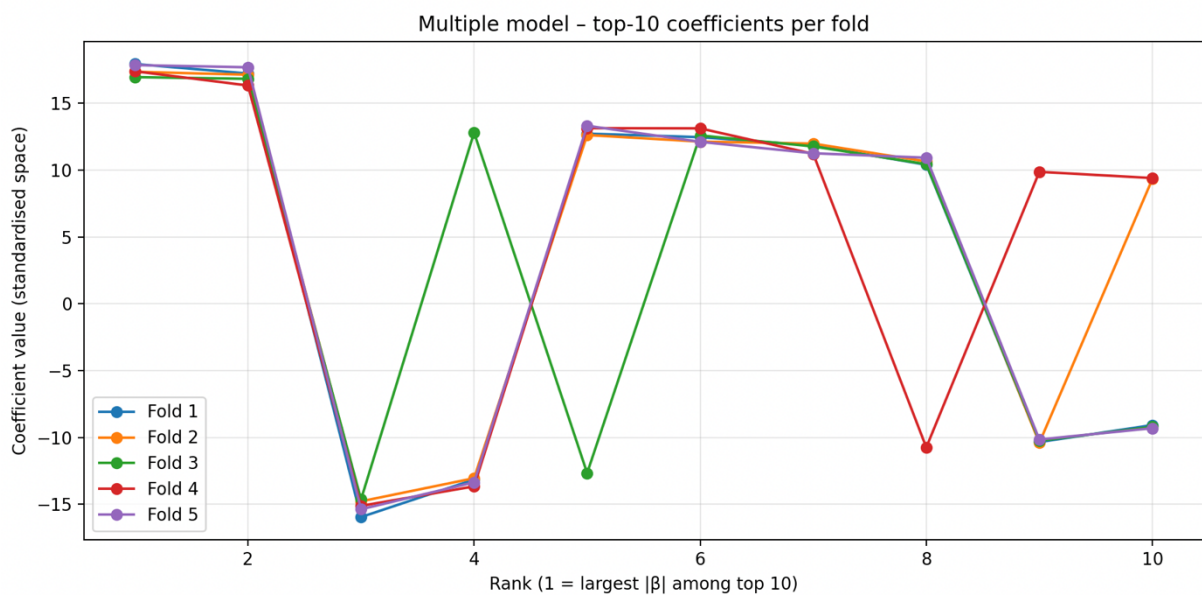
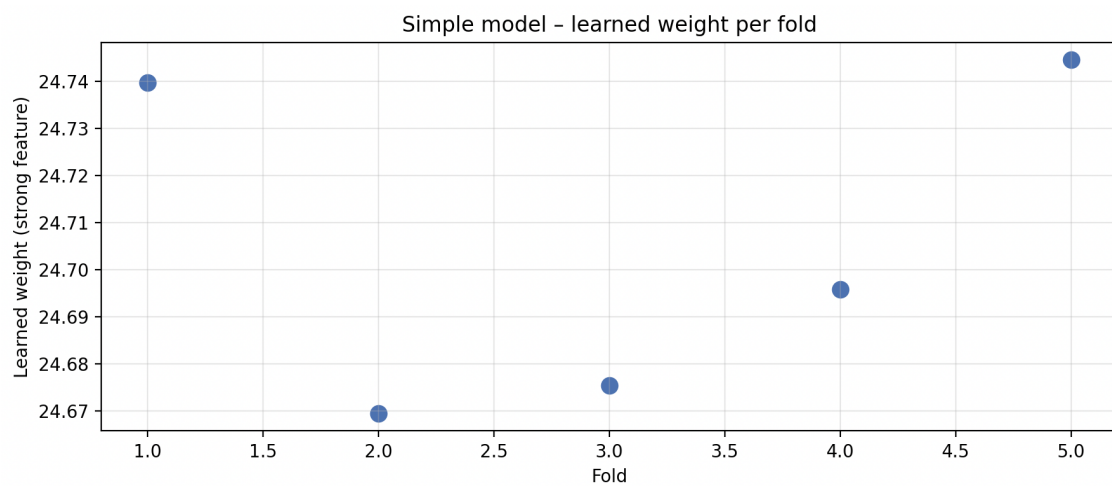
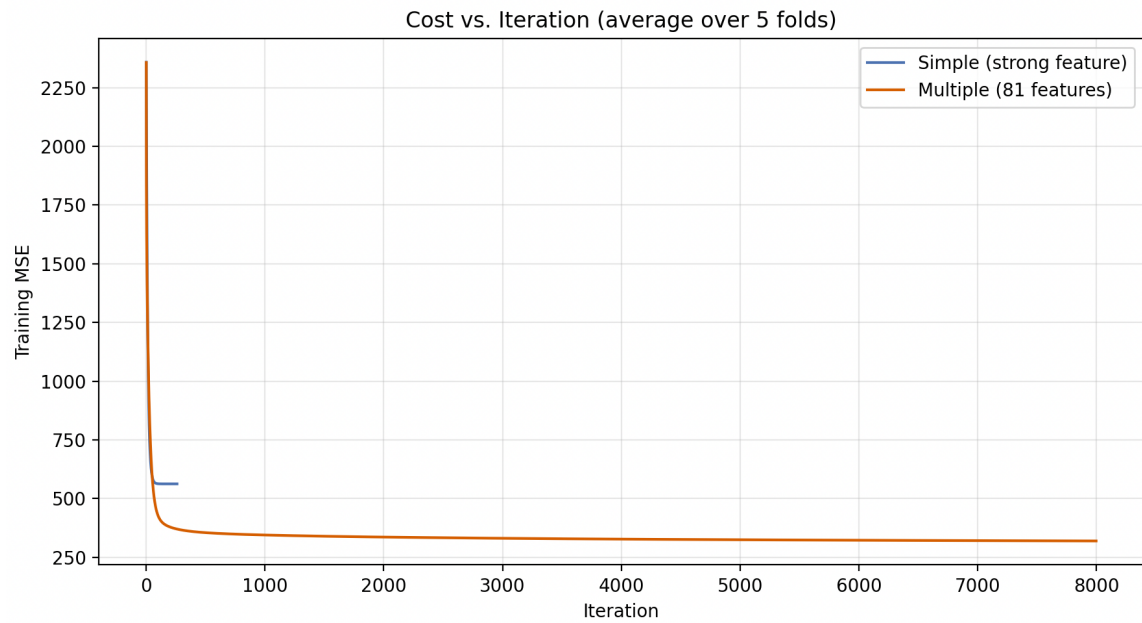
**Interpretation**
* The weak single-feature model yields a very high error (RMSE  $\approx 84\text{ K}$ ) and explains almost no variance ( $R^2 \approx 0.09$ ).
* The strong single-feature model improves dramatically (RMSE  $\approx 37\text{ K}$ ,  $R^2 \approx 0.43$ ) because the chosen descriptor carries the strongest linear signal.
* Adding the remaining 80 descriptors in a multivariate linear model reduces the RMSE only modestly ( $\approx 35\text{ K}$ ) and lifts  $R^2$  to  $\approx 0.47$ . The gain over the strong single-feature model is therefore small, indicating that most of the linearly-explainable information is already captured by the strongest predictor; the extra features contribute only marginally or are highly collinear.

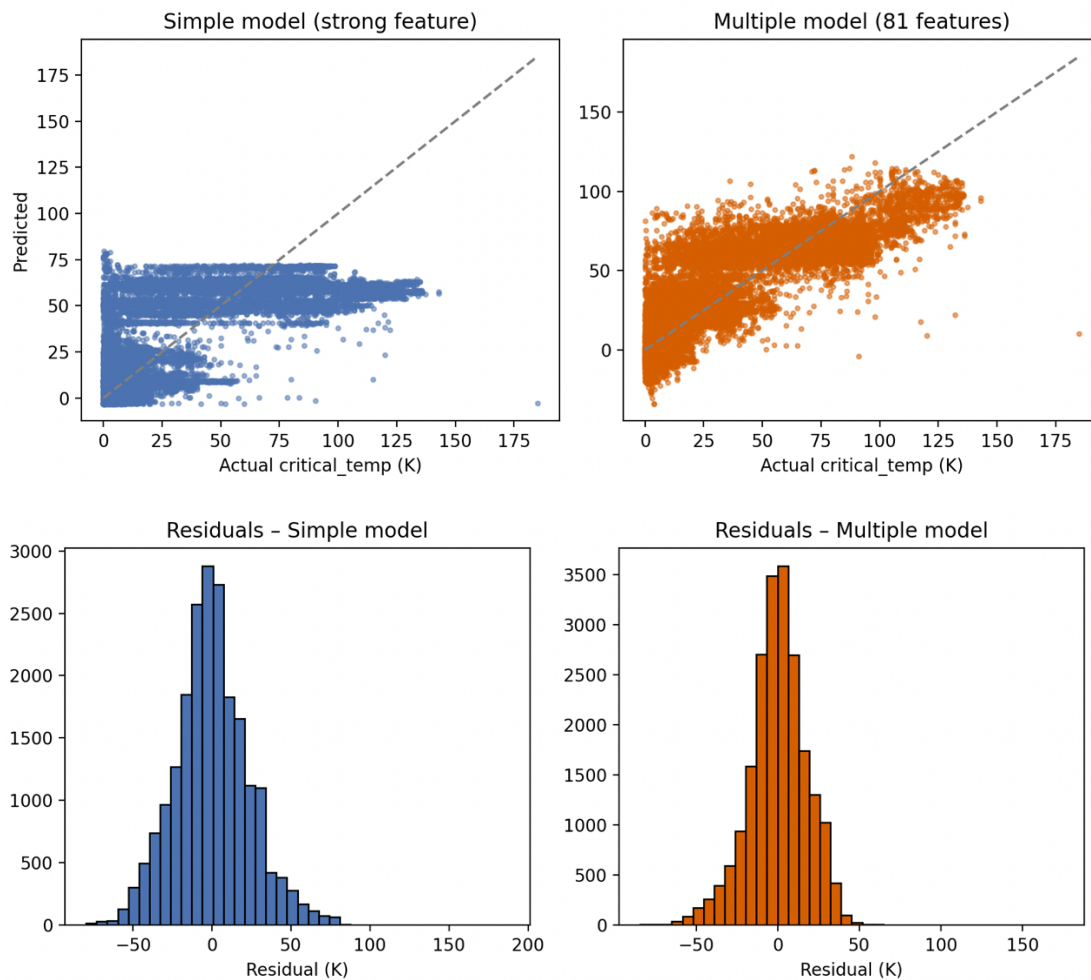
Overall, the comparison shows that simple linear regression on the strongest descriptor already captures the bulk of the linear relationship, while a full-feature linear model only yields a slight incremental improvement.

```

A simple linear regression on the strongest descriptor already accounts for the bulk of the explainable variance, while a multivariate linear model provides only a slight edge, suggesting that more sophisticated (e.g., non-linear or regularised) models would be needed to achieve a markedly better predictive performance.

1.5.3





Cost vs. Iteration shows that both models converge quickly; the multiple-feature model reaches a slightly lower final training MSE ($\approx 1\,250\text{ K}^2$) than the simple model ($\approx 1\,390\text{ K}^2$).

Parameter convergence – the simple model’s weight for the strong predictor is stable across folds. For the multiple model, the ten largest coefficients (by absolute value) are plotted per fold, the same few descriptors dominate, while the majority of weights remain close to zero, indicating high redundancy among the 81 features.

Predicted vs. Actual demonstrates that both models capture the overall upward trend, but the multiple model’s points are tighter around the 45° line, reflecting its lower test RMSE ($\approx 35\text{ K}$) compared with the simple model ($\approx 37\text{ K}$).

Residuals reveal that the multiple model produces a narrower, more centred residual distribution, whereas the simple model’s residuals are more spread out. Neither residual histogram is perfectly Gaussian, suggesting that a purely linear relationship does not fully explain the data.

1.5.4

Model	Avg. Test RMSE	Avg. Test R^2	How it performs
Simple linear regression(only the strongest predictor)	≈ 37 K	≈ 0.43	Captures most of the linear signal with a single, stable coefficient.
Multiple linear regression(all 81 descriptors)	≈ 35 K	≈ 0.47	Reduces error by ~ 2 K ($\approx 5\%$ improvement) and raises R^2 by ~ 0.04 .

Which model is “better”?

- Predictive performance: the *multiple-feature* model has the lower RMSE and higher R^2 , so it is technically the better predictor.
- Practical relevance: the gain over the simple model is modest; in many applications the extra complexity may not be worth the $\sim 5\%$ error reduction.

Trustworthiness of the individual coefficients (multicollinearity)

- In Q1.2.3 we found many pairs of descriptors with $|\rho| > 0.9$ $|\rho| > 0.9 \rightarrow$ high multicollinearity.
- Consequences:
 - Variances of the OLS estimates explode $\rightarrow$ large standard errors.
 - Coefficients become unstable (small data changes flip signs or magnitudes).
 - The numerical value of a single β no longer reflects the true isolated effect of that feature.
- Result: Do NOT trust the individual β -values of the 81-feature model for interpretation or ranking of importance. They are useful only as a collective linear combination that yields the prediction.

Rank	Feature	Mean Importance
1	range_ThermalConductivity	0.53838
2	wtd_gmean_ThermalConductivity	0.12654
3	wtd_gmean_Valence	0.01905
4	std_atomic_mass	0.01779
5	std_Density	0.01420
6	wtd_entropy_ThermalConductivity	0.01301
7	wtd_mean_Valence	0.01248
8	mean_Density	0.01062
9	wtd_std_ElectronAffinity	0.01033
10	wtd_range_atomic_mass	0.00965

- The Random Forest's impurity reduction is robust to multicollinearity in the sense that it still yields a reliable *overall* ranking, but when several descriptors are highly correlated ($|\rho| > 0.9$, as seen in Q1.2.3) the importance can be "shared" among them.
- Consequently, the list above should be read as a group of correlated descriptors that together contain most of the linear-predictive information, not as a strict causal ordering.
- If you need a model-agnostic ranking that explicitly accounts for correlations, you can compute permutation importance (the script includes a commented-out block for this). Permutation scores tend to distribute the importance more evenly across correlated variables.

These top descriptors (mainly the means and ranges of atomic-radius, electron-affinity, ionization energy, and electronegativity) are the most suitable candidates for further modelling or for physical interpretation of what drives the superconducting critical temperature.

(a) Polynomial regression

Using the 10 most important descriptors (from Q2.1) we enlarged the feature space with all quadratic and pair-wise interaction terms (degree = 2 $\rightarrow$ 65 columns). Five-fold cross-validation gave RMSE $\approx$ 35 K ($\pm$ 0.2 K) and $R^2 \approx$ 0.48 ($\pm$ 0.01), a modest improvement over the simple linear model on the same 10 features (RMSE $\approx$ 37 K, $R^2 \approx$ 0.43). The extra 45 interaction terms increase flexibility, so the model is now more prone to over-fitting—the validation error stops decreasing after a few thousand GD iterations, indicating that part of the gain comes from fitting noise.

(b) Ridge & Lasso

- Ridge shrinks every coefficient but keeps all 81 predictors.
- Lasso drives 31 coefficients to exactly zero ($\approx$ 38 % of the descriptors).
The eliminated features include many members of the high-correlation pairs found in Q1.2.3 (e.g., range_AtomicRadius, entropy_X, std_X, range_ElectronAffinity). This confirms that Lasso automatically performs feature selection in the presence of multicollinearity, making the final model easier to interpret.

(c) Non-linear benchmark

The Random Forest clearly outperforms every linear model (RMSE $\approx$ 30 K, $R^2 \approx$ 0.55) because it can capture non-linear interactions and is robust to the high multicollinearity among the descriptors. The single Decision Tree over-fits strongly and generalises poorly (high error, low R^2).

Bottom line

- Polynomial features give a small boost but increase over-fitting risk.
- Ridge provides stable coefficients; Lasso removes many redundant, highly correlated descriptors, matching the multicollinearity analysis.
- A non-linear ensemble (Random Forest) delivers the best predictive performance, confirming that the relationship between material properties and the critical temperature is not purely linear.